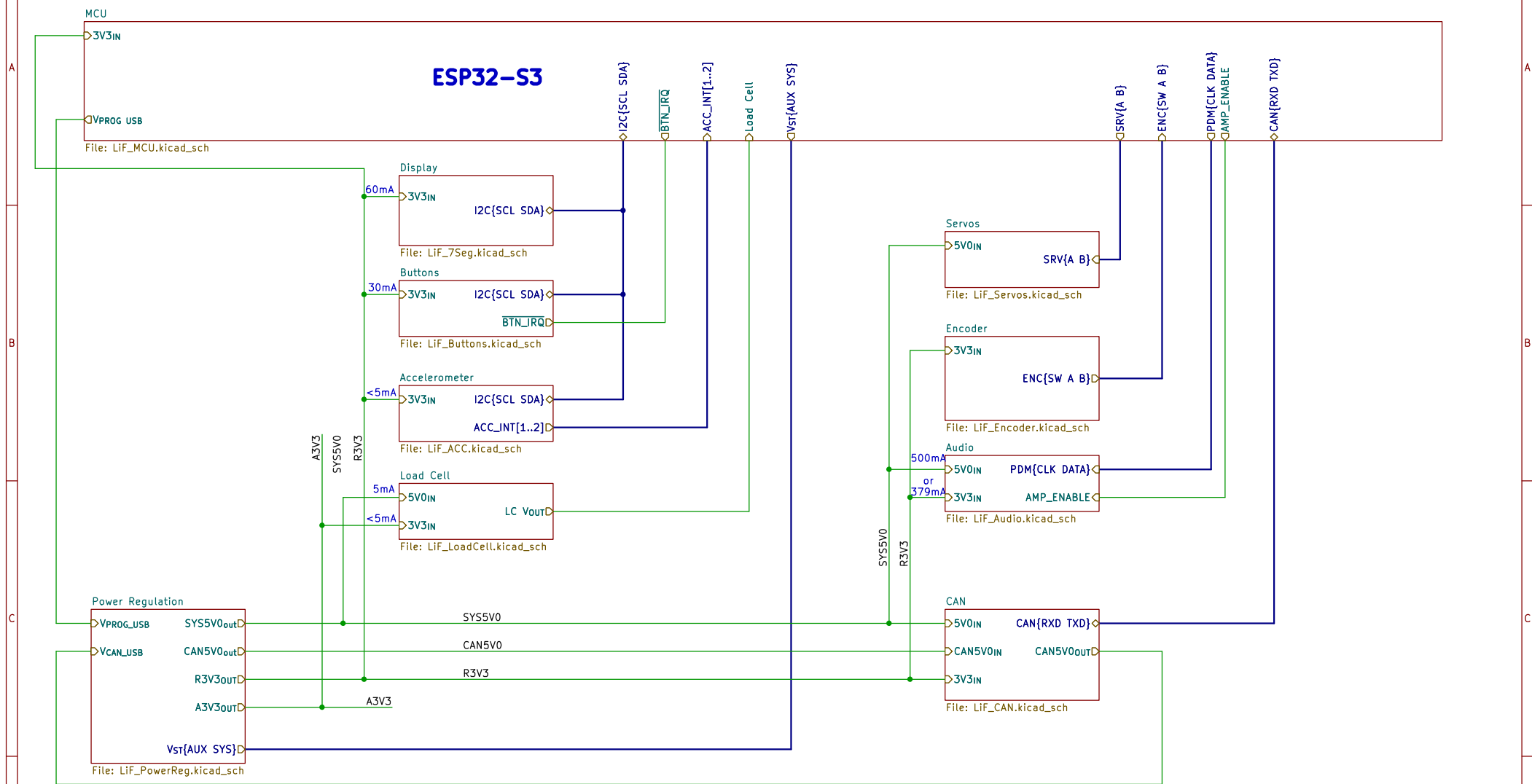


Lithium Firefly – Custom FC/CC PCB



Verified by MR
Designed by MR
Lithium Firefly Team

Sheet: /
File: LIF_FC-PCB_rev1.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-13
KiCad E.D.A. 9.0.4

Rev: 1
Id: 1/14

Lithium Firefly – Power Regulation

Aux V Supply

5V Switching Regulator

Power OR-ing

Digital 3V3 Linear Regulator

Analog 3V3 Linear Regulator

Vst.SYS
Vst.AUX

Vst{AUX SYS}

Verified by MR
Designed by MR
Lithium Firefly Team
Sheet: /Power Regulation/
File: LiF_PowerReg.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4	Date: 2026-07-12	Rev: 1
KiCad E.D.A. 9.0.4		Id: 2/14

Lithium Firefly Team

Lithium Firefly Team

Sheet: /Power Regulation/
File: LiF_PowerReg.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

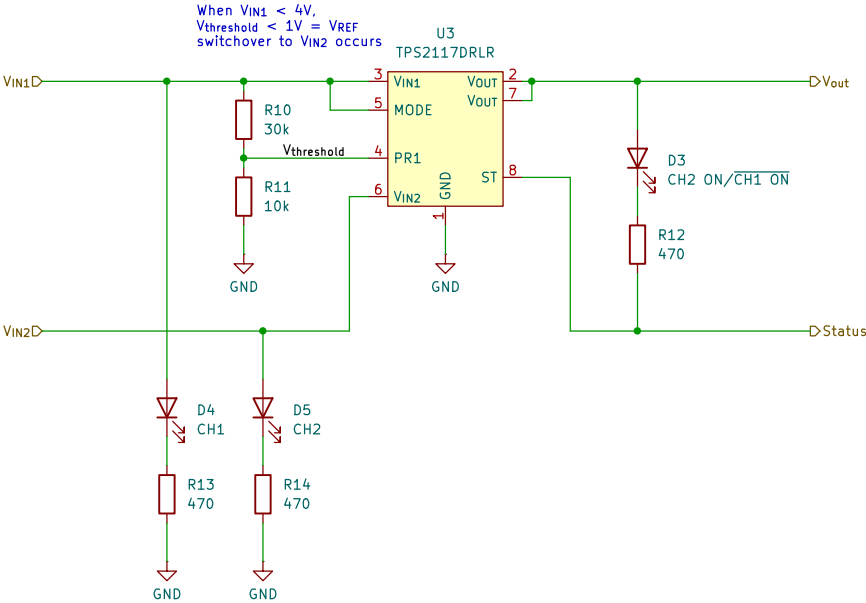
Size: A4	Date: 2026-07-12
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KiCad E.D.A. 9.0.4

Rev: 1

Id: 2/14

Lithium Firefly – Power OR-ing



Verified by MR
Designed by MR

Lithium Firefly Team

Sheet: /Power Regulation/Power OR-ing/
File: LiF_Power_OR-ing.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4

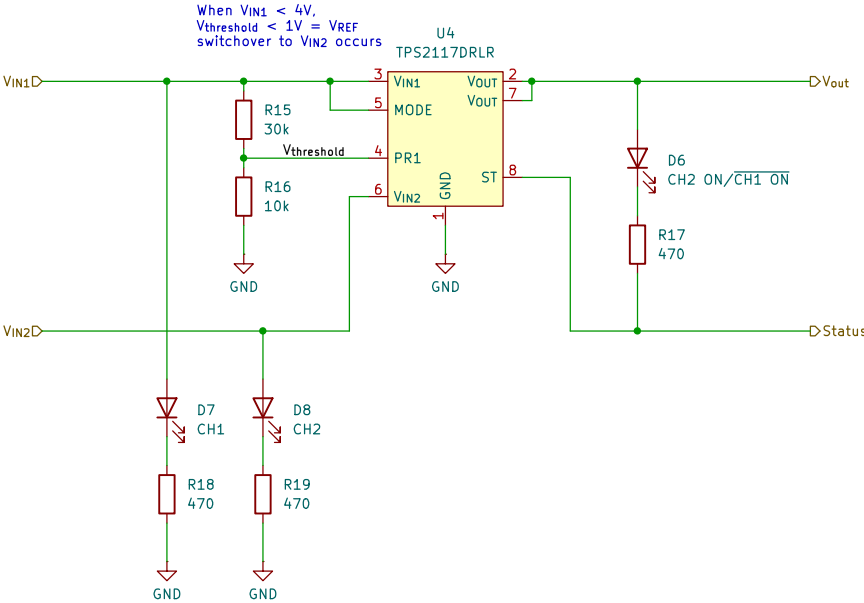
Date: 2026-07-11

Rev: 1

KiCad E.D.A. 9.0.4

Id: 3/14

Lithium Firefly – Power OR-ing



Verified by MR

Designed by MR

Lithium Firefly Team

Sheet: /Power Regulation/Power OR-ing1/

File: LiF_Power_OR-ing.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4

Date: 2026-07-11

Rev: 1

KiCad E.D.A. 9.0.4

Id: 4/14

[illegible]

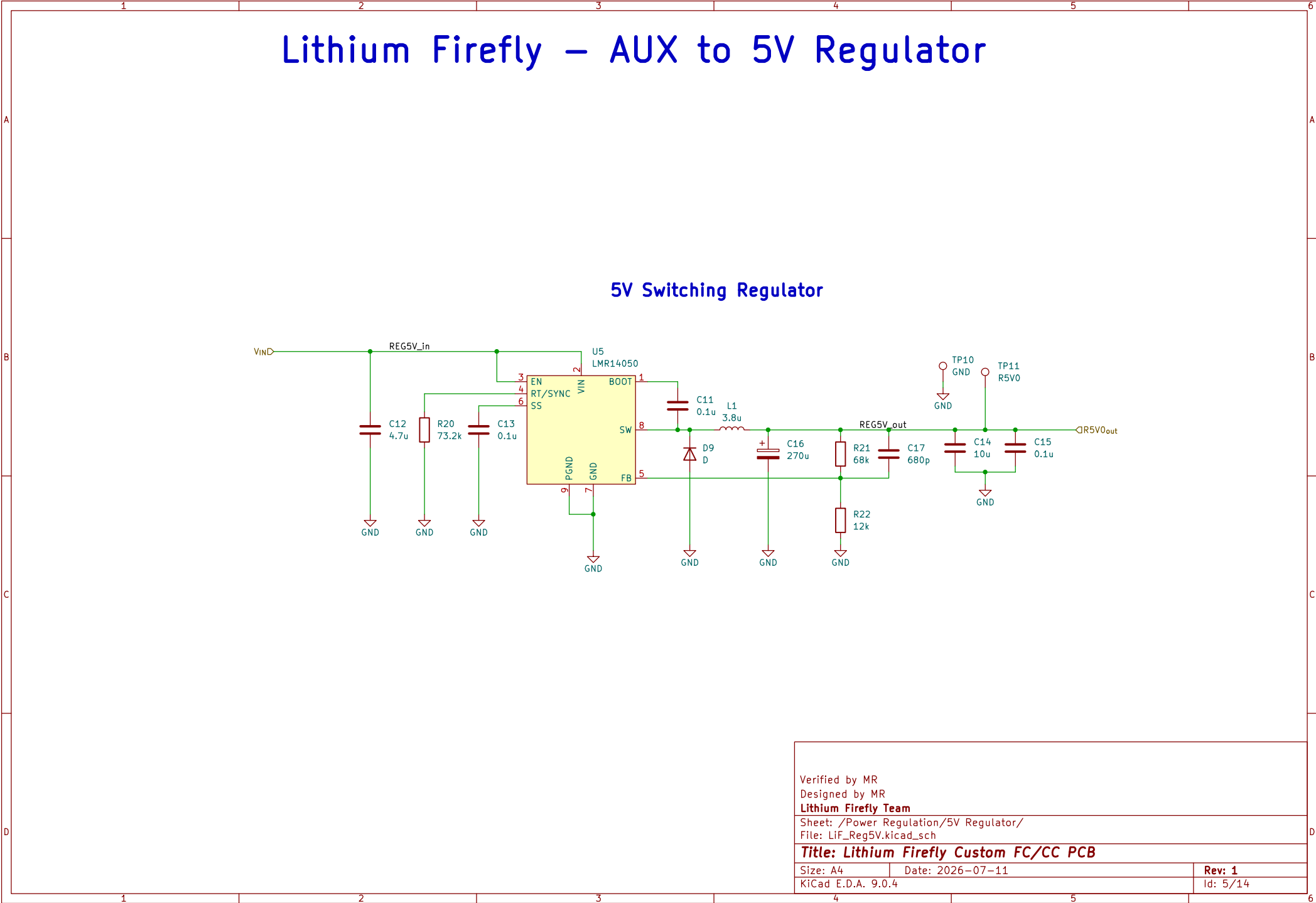
Lithium Firefly – AUX to 5V Regulator

5V Switching Regulator

The schematic diagram illustrates a 5V switching regulator circuit. The input voltage, labeled **VIND**, is connected to the **REG5V_in** node. This node is connected to the **VIN** pin (pin 2) of the **U5 LMR14050** IC. The **EN** pin (pin 3) is connected to **REG5V_in** through a 4.7uF capacitor (**C12**). The **RT/SYNC** pin (pin 4) is connected to **REG5V_in** through a 73.2k resistor (**R20**). The **SS** pin (pin 6) is connected to **REG5V_in** through a 0.1uF capacitor (**C13**). The **PGND** pin (pin 9) is connected to ground. The **GND** pin (pin 7) is connected to ground. The **BOOT** pin (pin 1) is connected to the **SW** node (pin 8) through a 0.1uF capacitor (**C11**). The **SW** node is connected to the **FB** pin (pin 5) through a 3.8uH inductor (**L1**). The **FB** pin is connected to ground through a 270uF capacitor (**C16**). The **FB** pin is also connected to the **REG5V_out** node through a 68k resistor (**R21**). The **REG5V_out** node is connected to ground through a 12k resistor (**R22**). The **REG5V_out** node is also connected to the **TP10 GND** node. The **REG5V_out** node is connected to the **TP11 R5V0** node. The **REG5V_out** node is connected to the **R5V0_out** node. The **R5V0_out** node is connected to ground through a 10uF capacitor (**C14**). The **R5V0_out** node is also connected to ground through a 0.1uF capacitor (**C15**). The **R5V0_out** node is connected to the **TP11 R5V0** node.

Verified by MR
Designed by MR
Lithium Firefly Team
Sheet: /Power Regulation/5V Regulator/
File: LiF_Reg5V.kicad_sch

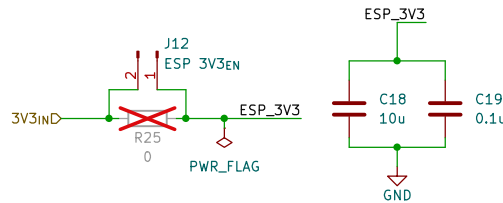
Title: Lithium Firefly Custom FC/CC PCB	
Size: A4	Date: 2026-07-11
KiCad E.D.A. 9.0.4	Rev: 1
Id: 5/14	



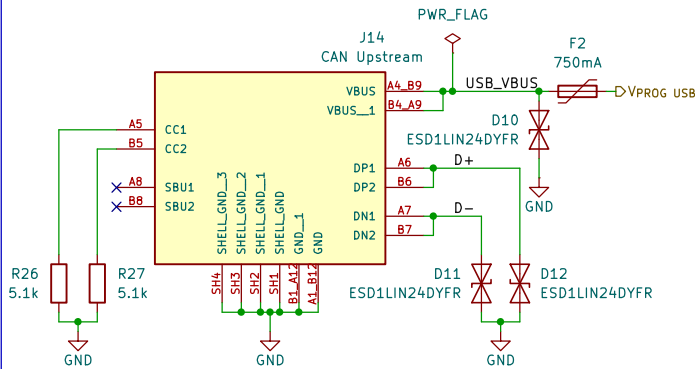
Id: 5/14

Lithium Firefly – MCU – ESP32–S3

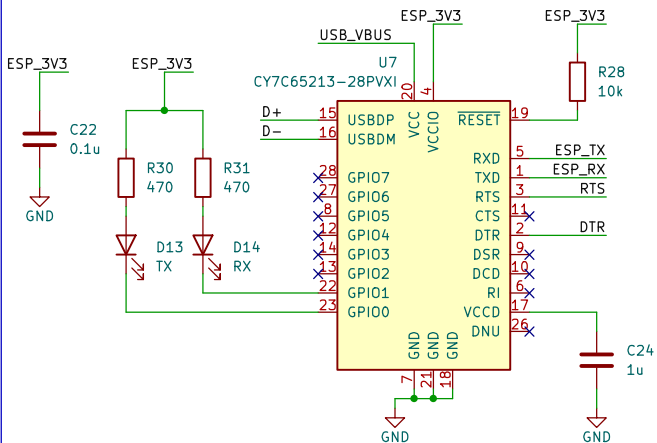
Power Input



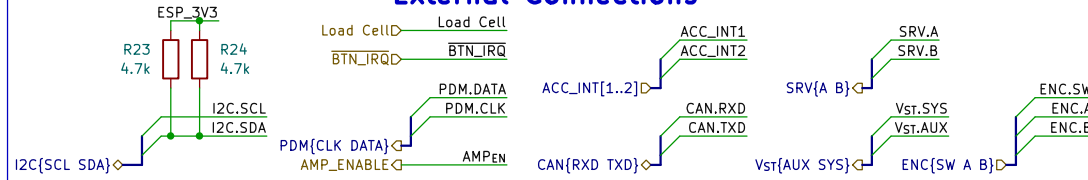
Programming USB



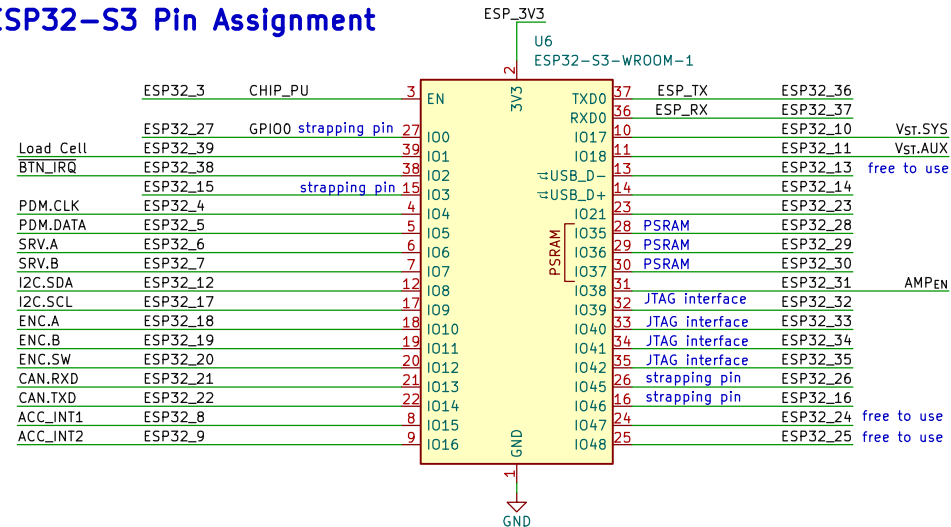
USB to UART bridge



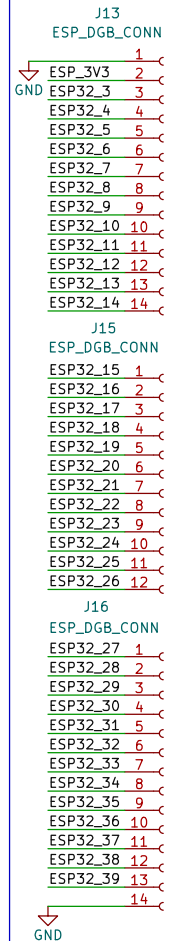
External Connections



ESP32–S3 Pin Assignment

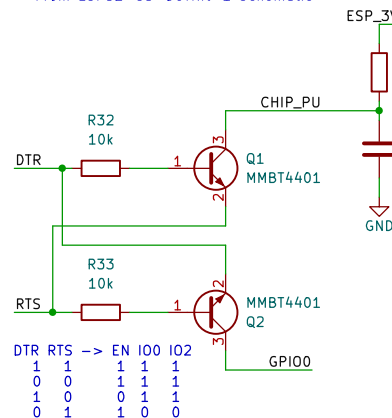


DEBUG CONNECTOR

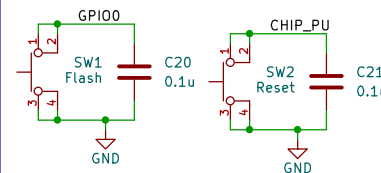


Auto Program Circuit

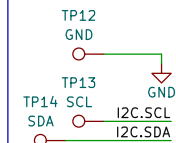
From ESP32–S3–DevKit–1 Schematic



Reset and Flash Buttons



I2C DBG



Verified by MR
Designed by MR
Lithium Firefly Team

Sheet: /MCU/
File: LiF_MCU.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

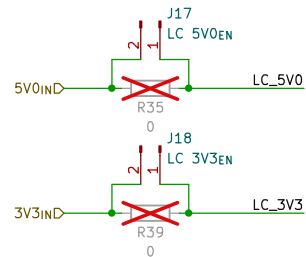
Size: A4 Date: 2026–07–13

KiCad E.D.A. 9.0.4

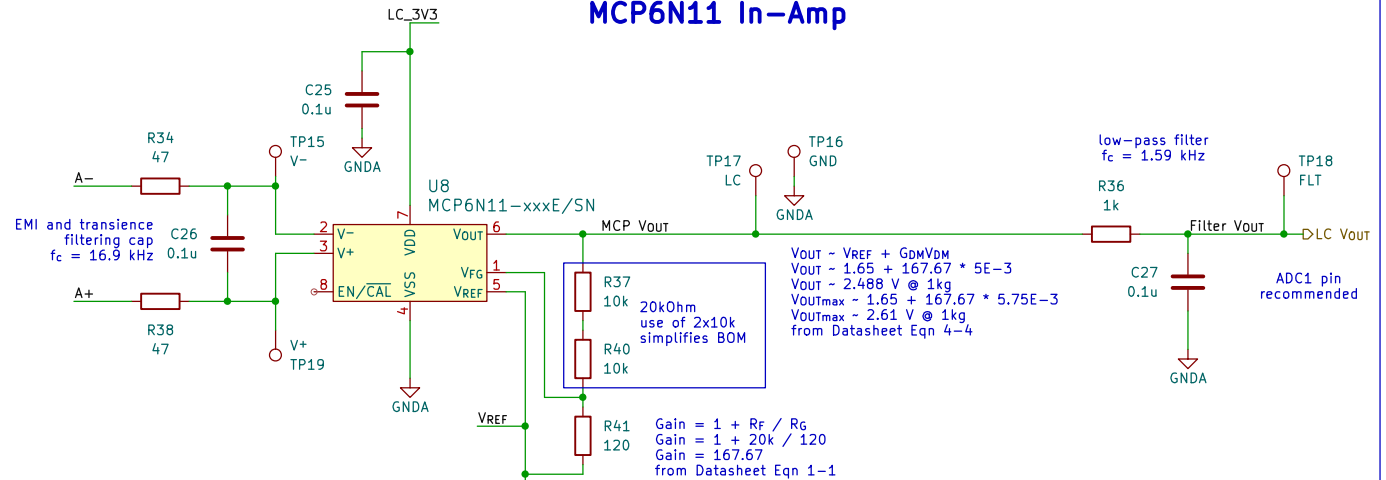
Rev: 1
Id: 6/14

Lithium Firefly – Load Cell Signal Conditioning

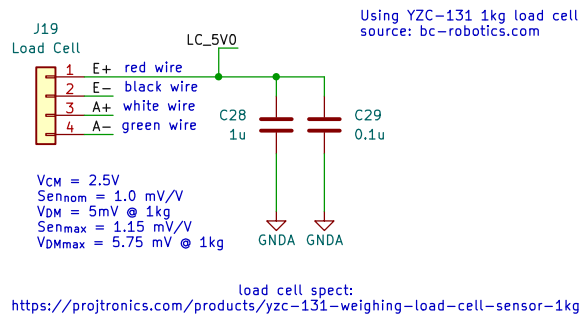
Power Input



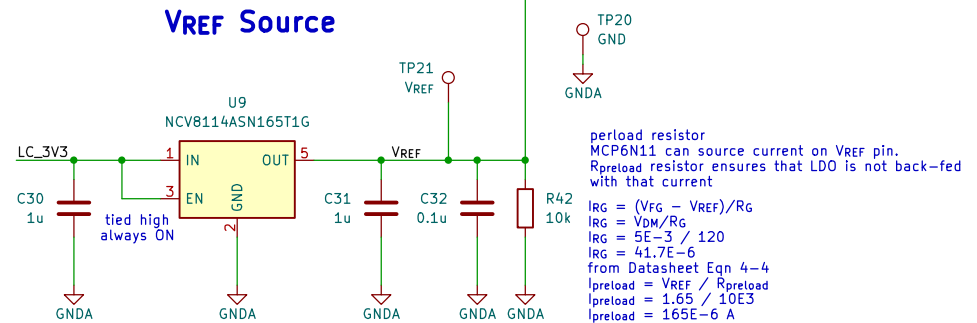
MCP6N11 In-Amp



Load Cell Connector



VREF Source



Verified by MR
Designed by RY and MR
Lithium Firefly Team

Sheet: /Load Cell/
File: LiF_LoadCell.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-11

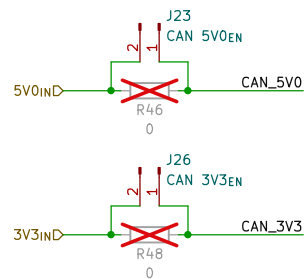
KiCad E.D.A. 9.0.4

Rev: 1

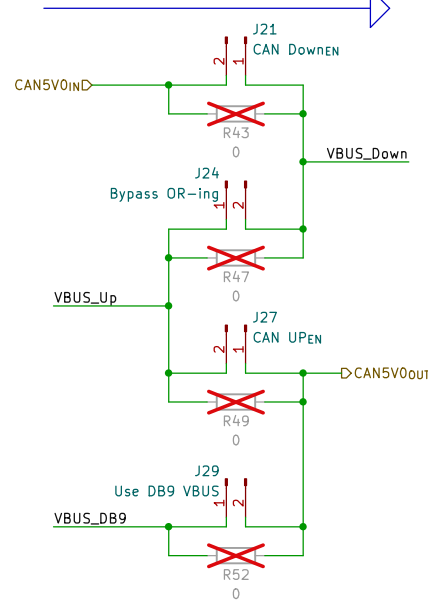
Id: 7/14

Lithium Firefly – CAN Transceiver and Connectors

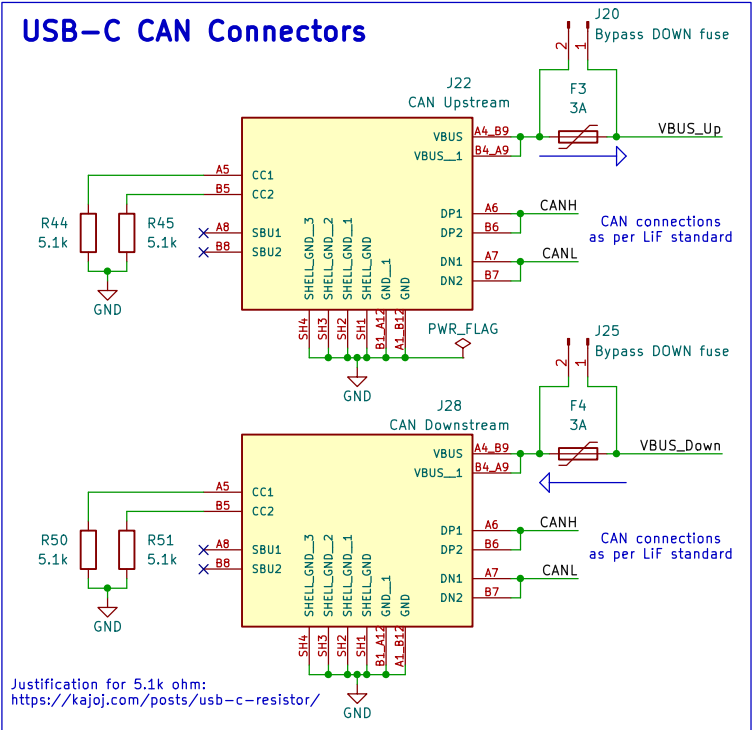
Power Input



Power Path Settings

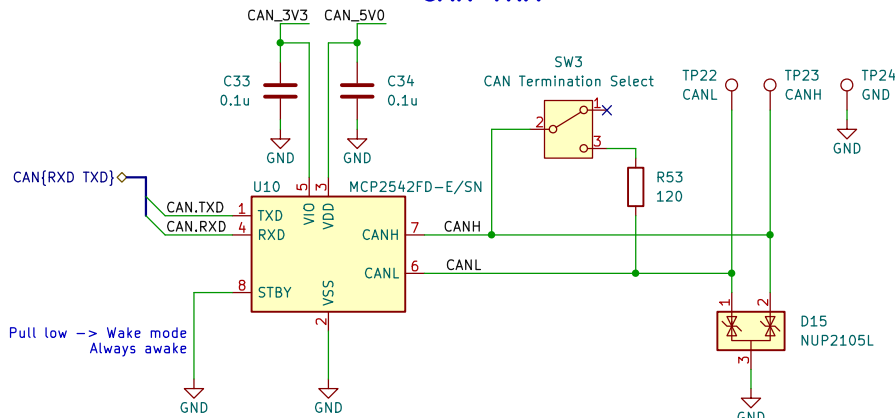


USB-C CAN Connectors

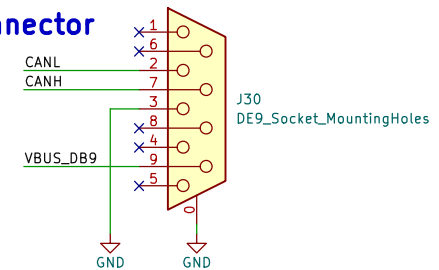


Justification for 5.1k ohm:
<https://kajo.com/posts/usb-c-resistor/>

CAN TRX



DB-9 CAN Connector



Verified by MR
 Designed by RY and MR
Lithium Firefly Team

Sheet: /CAN/
 File: LiF_CAN.kicad_sch

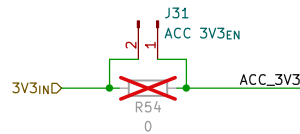
Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-11
 KiCad E.D.A. 9.0.4

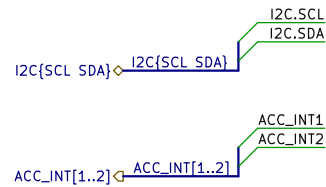
Rev: 1
 Id: 8/14

Lithium Firefly – Accelerometer

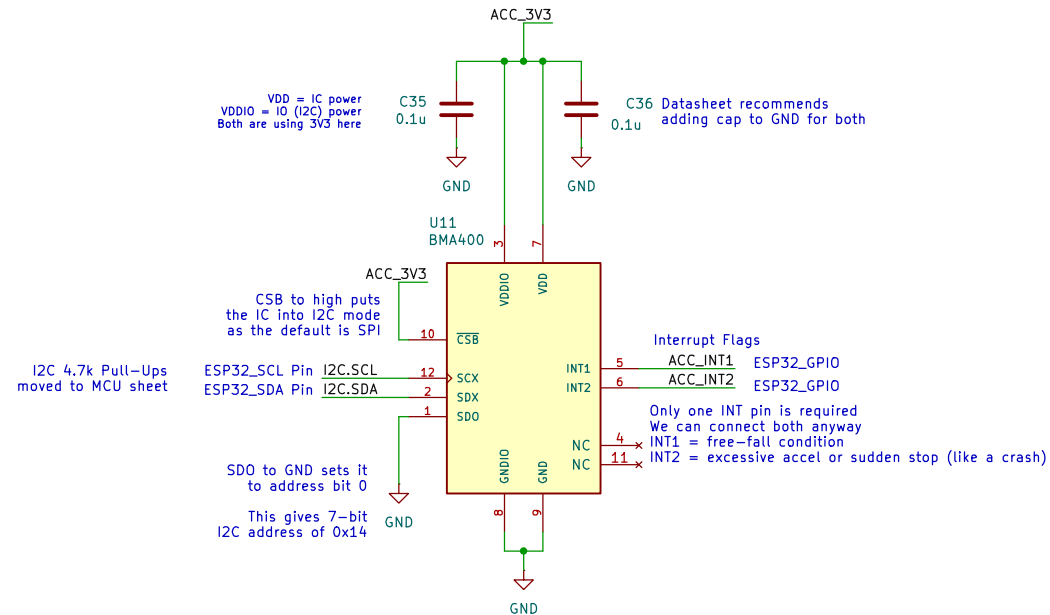
Power Input



Inputs



Accelerometer IC



Verified by MR
Designed by NK

Lithium Firefly Team

Sheet: /Accelerometer/
File: LiF_ACC.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-12

KiCad E.D.A. 9.0.4

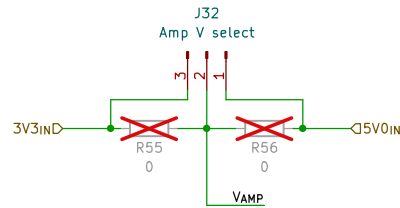
Rev: 1
Id: 9/14



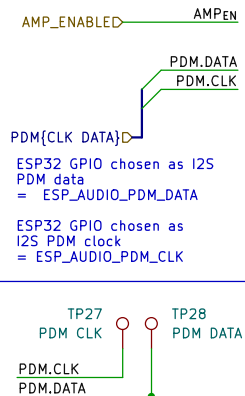
Lithium Firefly – Audio Amplifier

ESP Audio out -> double RC filter (tuned at 15 kHz) -> DAC -> amplifier -> speaker

Power Source Selection



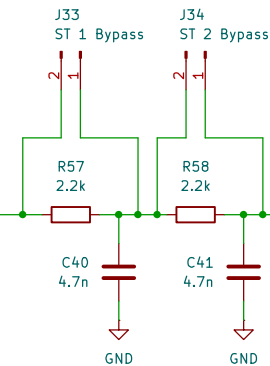
Inputs



Low-Pass Filter

LPF filter setup as the I2S can use PDM (pulse data modulation)
4.7nF and 2.2kOhm produce an Fc of 15.4 kHz.
Check NK logbook (2026-07-10) for more detail.

A 1.8 nF can be used as well to give a cutoff frequency of 40 kHz cutoff very high frequency noise but allow higher pitches (like 15k + lol)



LPF to let low frequency pass while cutting off high switching frequency from ESPs pin

$$F_c = 1 / (2\pi * \sqrt{RC})$$

Let $F_c = 15$ kHz (to cut out high frequency noise; 15 kHz is good enough)

Let's find values.

Let $C = 4.7$ nF, find R

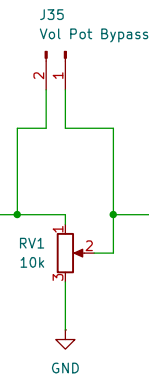
Using the above formula,

$R = 2.2$ k (at least for common values)

And the final cutoff frequency is 15.4 kHz

Volume Adjustment

Variable gain (limit output current from filter that is entering the IC to reduce gain)



AC-Coupling

IC has internal input R of 10k

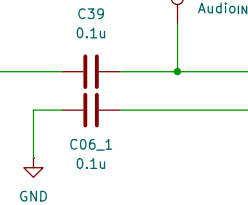
Using 100 nF cap gives:

$$F_c = 1 / ((2\pi)(RC))$$

$$F_c = 1 / ((2\pi)(100n * 10k))$$

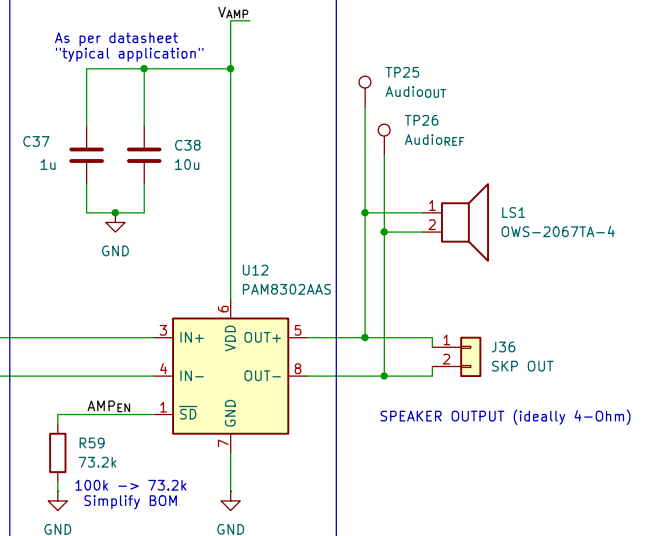
$$F_c = 160 \text{ Hz}$$

This is DC cutoff Freq (hence the caps).



Amplifier IC

As per datasheet "typical application"



Verified by MR

Designed by NK

Lithium Firefly Team

Sheet: /Audio/

File: LiF_Audio.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4

Date: 2026-07-12

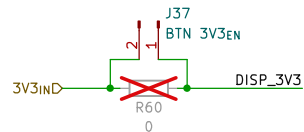
Rev: 1

KiCad E.D.A. 9.0.4

Id: 10/14

Lithium Firefly – 7-Segment Display and Driver

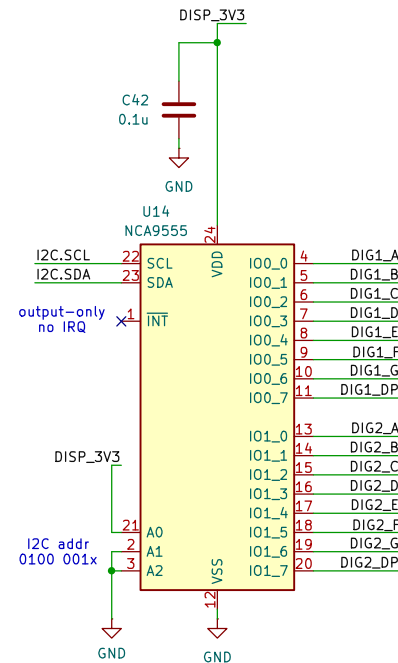
Power Input



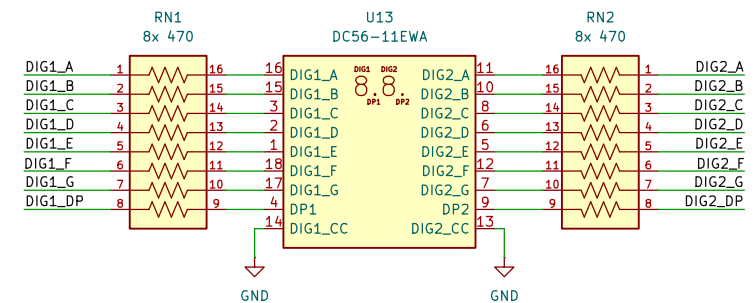
Inputs



I2C GPIO Expander



7-Segment Display



Verified by MR
Designed by NK and MR
Lithium Firefly Team

Sheet: /Display/
File: LiF_7Seg.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-11

KiCad E.D.A. 9.0.4

Rev: 1
Id: 11/14



Lithium Firefly – Button and LED Array

Power Input

Inputs/Outputs

Buttons and Debounce

I2C GPIO Expander

Floor Selection LEDs

Verified by MR
Designed by NK and MR

Lithium Firefly Team

Sheet: /Buttons/

File: LiF_Buttons.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4

Date: 2026-07-11

Rev: 1

KiCad E.D.A. 9.0.4

Id: 12/14

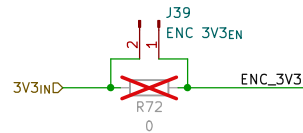


Rev: 1
Id: 12/14

Lithium Firefly – Rotary Encoder

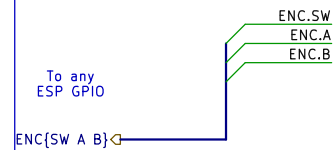
ENCODER MODULE – FLOOR SELECTOR

Power Input

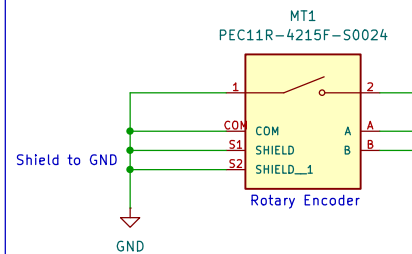


Also requires BTN 3V3EN to be ON

Inputs/Outputs



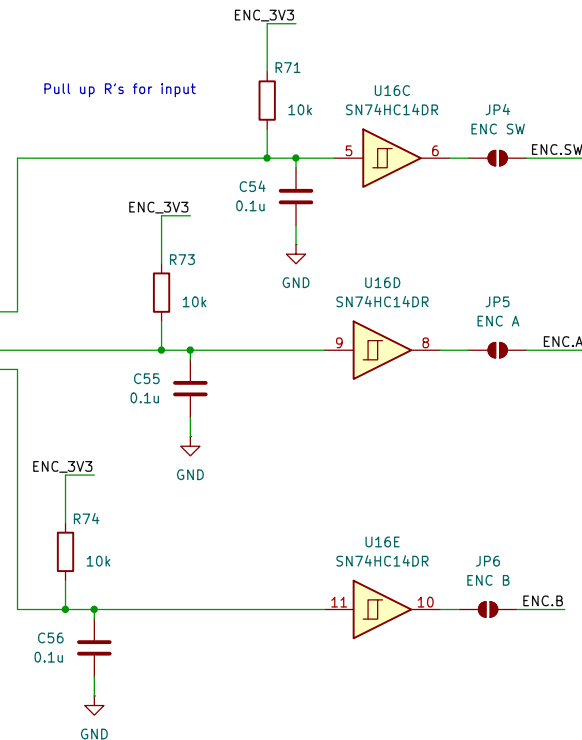
Encoder



Function (car controller)
Encoder will be rotated and the value
will be updated on the 7-seg display.
This will behave as a floor selector

Delete if not wanted :(

Debounce



Verified by MR
Designed by NK
Lithium Firefly Team

Sheet: /Encoder/
File: LiF_Encoder.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-11

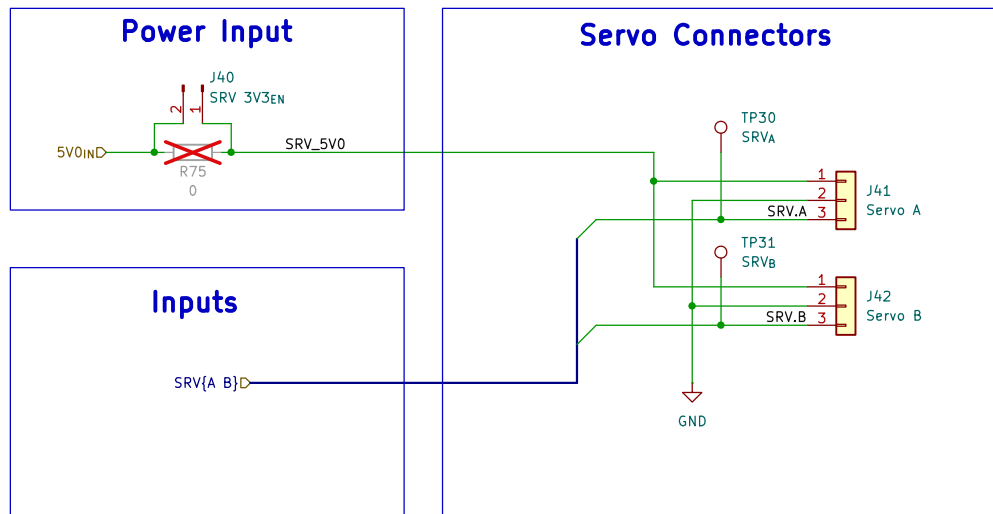
KiCad E.D.A. 9.0.4

Rev: 1

Id: 13/14



Lithium Firefly – Servo Connectors



Verified by MR
Designed by MR
Lithium Firefly Team

Sheet: /Servos/
File: LiF_Servos.kicad_sch

Title: Lithium Firefly Custom FC/CC PCB

Size: A4 Date: 2026-07-12

KiCad E.D.A. 9.0.4

Rev: 1

Id: 14/14